In [1]:
```
import os
os.getcwd()
```

Out[1]:
```
'C:\\Users\\Pranav\\Python Assignment'
```

In [8]:
```
#1.Read a file containing student names (one per line) and store them in a list.
f = open("names.txt", "r")
names = f.readlines()
f.close()
names = [name.strip() for name in names]
print(names)
```

```
['Pranav', 'Aayush', 'Gaurav', 'Rudra', 'Rohan', 'Prasad', 'Prachi', 'Shrusti']
```

In [9]:
```
#2.Print names longer than 5 characters.
f = open("names.txt", "r")
names = f.readlines()
f.close()
names = [name.strip() for name in names]

for name in names:
    if len(name) > 5:
        print(name)
```

```
Pranav
Aayush
Gaurav
Prasad
Prachi
Shrusti
```

In [10]:
```
#3.Count how many names start with a vowel
f = open("names.txt", "r")
names = f.readlines()
f.close()
names = [name.strip() for name in names]
count = 0
for name in names:
    if name[0].lower() in "aeiou":
        count += 1
print("Number of names started with vowels are:", count)
```

```
Number of names started with vowels are: 1
```

In [16]:
```
#4.Save the filtered names (longer than 5 characters) into a new file.
f = open("names.txt", "r")
f.close()
names = [name.strip() for name in names]
f1 = open("f_names.txt", "w")
for name in names:
    if len(name) > 5:
        f1.write(name + "\n")
f.close()
print("Names longer than 5 characters are saved into new file")
```

```
Names longer than 5 characters are saved into new file
```

In [18]:
```python
#5.Sort the list of names alphabetically and print them in ascending order.
f = open("names.txt", "r")
names = [name.strip() for name in names]
f.close()

names.sort()

print("Names in ascending order")
print(names)
```

```
Names in ascending order
['Aayush', 'Gaurav', 'Prachi', 'Pranav', 'Prasad', 'Rohan', 'Rudra', 'Shrusti']
```

In [26]:
```python
#6.Print names that contain the letter "a".
f = open("names.txt", "r")
names = [name.strip() for name in f]
f.close()

for name in names:
    if "a" in name.lower():
        print(name)
```

```
Pranav
Aayush
Gaurav
Rudra
Rohan
Prasad
Prachi
```

In [28]:
```python
#7.Count word frequency in names (handle duplicates) and store results in a diction
f = open("names1.txt", "r")
names = [name.strip() for name in f]
f.close()

frequency = {}

for name in names:
    if name in frequency:
        frequency[name] += 1
    else:
        frequency[name] = 1
print(frequency)
```

```
{'Pranav': 2, 'Aayush': 1, 'Gaurav': 1, 'Rudra': 1, 'Rohan': 2, 'Prasad': 1, 'Prach
i': 1, 'Shrusti': 3}
```

In [31]:
```python
#8. String Analysis
# o For each name, print:
# ▪ Length of the name
# ▪ First and last character
# ▪ Whether it is a palindrome
f = open("names.txt", "r")
names = [name.strip() for name in f]
f.close()
```

```
for name in names:
    print("Length",len(name))
    print("First Character",name[0])
    print("Last Character",name[-1])

    if name.lower() == name [::-1].lower():
        print("Palindrome")
    else:
        print("Not Palindrome")
print()
```

```
Length 6
First Character P
Last Character v
Not Palindrome
Length 6
First Character A
Last Character h
Not Palindrome
Length 6
First Character G
Last Character v
Not Palindrome
Length 5
First Character R
Last Character a
Not Palindrome
Length 5
First Character R
Last Character n
Not Palindrome
Length 6
First Character P
Last Character d
Not Palindrome
Length 6
First Character P
Last Character i
Not Palindrome
Length 7
First Character S
Last Character i
Not Palindrome
```

In [37]:
```
#9.File + List + String Combined Challenge
# o Read names from file → list.
# o Remove duplicates.
# o Sort by length (shortest to longest).
# Write results into processed_names.txt

f = open("names1.txt", "r")
names = [name.strip() for name in f]
f.close()
```

```
u_names = list(set(names))
u_names.sort(key=len)

f = open("Processed_names.txt", "w")

for name in u_names:
    f.write(name + "\n")

f.close()

print("Names stored in new file successfully")
print(u_names)
```

```
Names stored in new file successfully
['Rudra', 'Rohan', 'Prachi', 'Aayush', 'Gaurav', 'Prasad', 'Pranav', 'Shrusti']
```

In [42]:
```
#10.o Read names from file.
# o Group them by starting letter (e.g., all names starting with
# A together).
# o Save each group into separate files (A_names.txt,
# B_names.txt, etc.).
from collections import defaultdict
groups = defaultdict(list)
for name in names:
    groups[name[0].upper()].append(name)
for letter, group in groups.items():
    file = open(letter + "_names.txt", "w")
    for name in group:
        file.write(name + "\n")
    file.close()
print("Completed")
```

```
Completed
```

In [ ]:
```

```